Does a JEPA world model help fly an unidentified fixed-wing aircraft?

A controlled test of the SkyJEPA recipe on a simulated Ultra Stick 120

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Answer in one line

In this setting, no. On a fixed wing whose commands are not forces, **JEPA + physics** reproduces SkyJEPA's ranking on training-range aircraft, but the controls SkyJEPA did not run predict better, the JEPA latent keeps little motor or CG information, and with equal noise training **JEPA + physics + noise** tracks worse under every noise kind.

1 The problem

Goal. Fly a fixed-wing aircraft whose aerodynamics, motor and center of gravity are unknown (new airframe, new payload or propeller) from its last measurements, with noisy sensors, without an identification flight.

What exists

- **System identification**: one dedicated test campaign per airframe, redone after every change.
- **Adaptive control** (L1 [9], INDI [10]): adapts the controller but gives no multi-step model to plan with.
- **Learned adaptation**: RMA [6] (policy trained with the true parameters), Neural-Fly [11] (learned residual forces with online adaptation).

Approach studied. A learned **world model inside MPPI** [4], which predicts how the current aircraft will move.

2 What SkyJEPA claims, and what it did not test

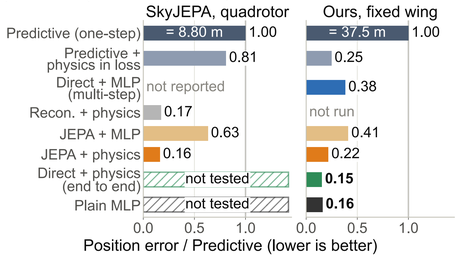
SkyJEPA [1]: a JEPA world model [3] of a quadrotor (TCN encoders, GRU predictor, SIGReg [2]) plus a physics-informed prober, trained in simulation on randomized vehicles, used in MPPI.

- Their Table III (horizon not stated): **JEPA + physics prober 1.43 m**, best; reconstruction + **physics prober 1.53 m**; **JEPA** + MLP prober **5.56 m**; Predictive **8.80 m**.
- With the **physics prober**, JEPA adds only **0.10 m** over reconstruction (1.43 vs 1.53 m).

Their actions are the four motor forces, so the thrust is known from the action:

$$\dot{v} = \frac{\sum_i f_i}{m} R\, e_3 - g + \Delta\dot{v}$$

Not tested: a plain MLP, and the physics prober on a predictive (state-loss) model, frozen or trained end to end.

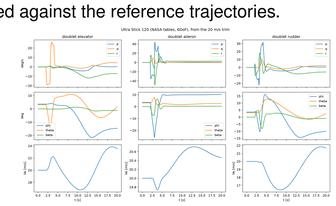


Position error divided by the one-step Predictive model (SkyJEPA: 8.80 m; ours: 37.5 m at 5 s, training-range aircraft, mean of 3 seeds).

Q1 Does JEPA predict better?
Q2 Does it identify the aircraft without labels?
Q3 Is it less sensitive to sensor noise at equal training?
Q4 Does it fly better in closed loop?

3 Test bench

Simulator. Python port of the University of Minnesota Ultra Stick 120 6-DoF model [8] (NASA wind-tunnel tables), checked against the reference trajectories.



Data. 2,000 training flights of 30 s in calm air (steady wind, no turbulence or gusts) and **200** test flights of 120 s at 20 Hz, flown by a TECS autopilot [12] with excitation maneuvers. Every flight has its own aircraft:

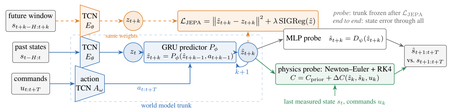
randomized	training	extreme*
mass	$\pm 20\,\%$	20–35 %
inertias	$\pm 30\,\%$	30–45 %
aero coefficients (6, each)	$\pm 20\,\%$	20–30 %
motor power	$\pm 30\,\%$	30–45 %
CG shift (of chord)	$\pm 3\,\%$	3–5 %

* half the test flights, all parameters at once

Commands are elevator, aileron, rudder, throttle: **not forces**. This is the key difference with SkyJEPA.

State: 9 quantities (airspeed, angle of attack, sideslip, p, q, r, roll, pitch, vertical speed).

Sizes as in SkyJEPA: state TCN [8,8,16], action TCN [4,4,8], GRU 24, $z \in \mathbb{R}^{24}$, SIGReg $\lambda = 0.02$, training horizon 20 steps; history 11 steps = 0.55 s (SkyJEPA: 10); the estimated wind is also an input.



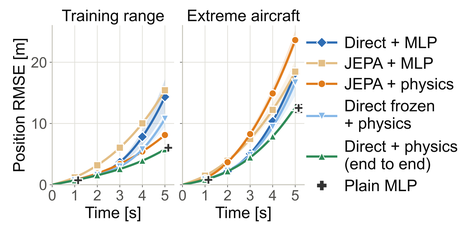
probe = head trained on the frozen trunk.

Models compared, 10.6–11.5k parameters (MLP 10,559; latent models 10,713; physics-head models 11,542):

this work	SkyJEPA name
Predictive (one-step)	Predictive
Direct + MLP (state loss over 1-s rollouts)	not reported
JEPA + MLP	JEPA + prober
JEPA + physics	JEPA + PI prober (their recipe)
Direct frozen + physics	not tested
Direct + physics (end to end)	not tested
Plain MLP: raw 0.55-s window, next state = last state + MLP [7]	not tested

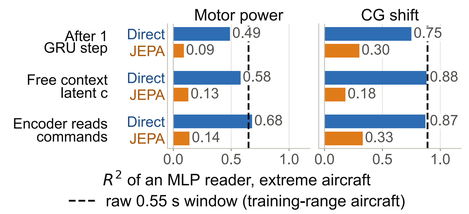
Physics head: outputs force and moment coefficients, is given the true mass and inertias (no aerodynamic prior), and integrates Newton–Euler with RK4 from the last measured state. The MLP-head models and the **Plain MLP** get no aircraft parameters.

4 Q1, open loop: SkyJEPA's ranking holds, the missing controls win



- Training-range aircraft, 5 s, 3 seeds: we reproduce SkyJEPA's ranking. **JEPA + physics 8.1 m** beats **JEPA + MLP 15.4 m**, **Direct frozen + physics 10.8 m** and **Predictive (one-step) 37.5 m**.
- The two controls SkyJEPA did not run do better: **Direct + physics (end to end) 5.8 m**, **Plain MLP 6.0 m**.
- Extreme aircraft: **JEPA + physics** is the worst, **23.6 m** (**JEPA + MLP** 18.5, **Plain MLP** 12.5, **Direct + physics (end to end)** 12.7).

5 Q2, identification: the JEPA latent drops the aircraft



R^2 of an MLP reader per aircraft parameter (learned representations on extreme aircraft, raw window on training-range aircraft).

- The raw 0.55-s window (states + commands) holds the motor (**0.65**) and the CG (**0.89**).
- **Direct + MLP** recovers them after one GRU step (**0.49** / **0.75**).
- **JEPA** does not (**0.09** / **0.30**), even when its encoder also reads the commands (**0.14** / **0.33**, before the GRU), or with a free context latent c that is not a prediction target (**0.13** / **0.18** vs **0.58** / **0.88** for the **Direct** trunk with c).
- Mass and C_L are read by both (about 0.5).

Why. In a JEPA the GRU memory is the latent itself, and the loss pins $\hat{z}_{t+k}$ to the encoding of a state window, which holds almost no motor or CG information ($R^2 \approx 0$). Why the free latent stays poor is open; one hypothesis: the objective favors slow, easy-to-predict content.

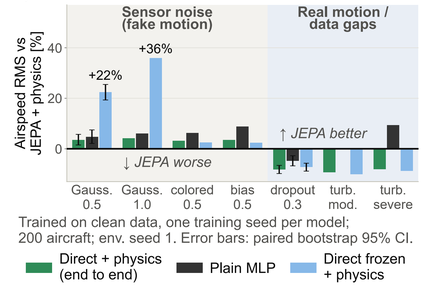
Consequence. A frozen **Direct** trunk + **physics head** matched end-to-end training only with a free context latent (training range / extreme):

Direct trunk + context, frozen, + physics, seed 0	**6.0** / **11.1 m**
Direct + physics (end to end), seed 0	**5.9** / **12.2 m**
Direct frozen + physics, no context, 3 seeds	**10.8** / **16.7 m**
JEPA trunk + context, frozen, + physics, seed 0	**8.0** / **20.6 m**

6 Q3, noise: JEPA helps only when nothing is trained with noise

Clean training (no model sees noise)

Open loop: **JEPA + MLP** starts high (**0.23** vs **0.045** for the **Plain MLP** at $\sigma = 0$) and grows less in relative terms ($\times 2.5$ vs $\times 10$ at $\sigma = 0.5$); it stays above the **Plain MLP** at every σ.

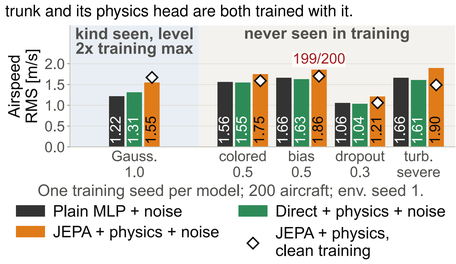


Closed loop (MPPI, 200 aircraft, environment seed never used for tuning, one training seed per model):

- **JEPA + physics** tracks airspeed **3–9 %** better than **Direct + physics (end to end)** and the **Plain MLP** under the four sensor-noise kinds (Gaussian 0.5: **1.19** vs **1.24** and **1.25 m/s**, paired bootstrap 95 % CI excludes 0; Gaussian 1.0: **1.67** vs **1.74** and **1.77**).
- It is worse than **Direct + physics (end to end)** under real motion and data gaps (severe turbulence **1.49** vs **1.37 m/s**; dropout 0.3: **1.05** vs **0.97**). It smooths, consistent with [5] (not tested directly).

Equal noise training

All three get the same augmentation: Gaussian noise (σ up to 0.5 of each state's std) on the past states, clean targets (for the JEPA, the target encoder sees the clean windows); the JEPA trunk and its physics head are both trained with it.



- **JEPA + physics + noise** tracks airspeed **12–27 %** worse than **Plain MLP + noise** and **Direct + physics + noise** in all five conditions, including the four noise kinds never seen in training. Gaussian 1.0 (kind seen in training, level twice the training maximum): **1.55** vs **1.22** and **1.31 m/s**.
- Open loop, on unseen bias and colored noise the three are within **0.02** (normalized 1-s error); **JEPA + physics + noise** is the least accurate at 5 s (**10.4 m** vs **6.6** and **7.2 m**).

Noise augmentation gives what the JEPA trunk gave, and more: under Gaussian 1.0 the **Plain MLP** goes from **1.77** (clean) to **1.22 m/s** (**Plain MLP + noise**).

7 Q4, closed loop: the Plain MLP and every physics-head model fly

Survivors out of 200	Gauss. 0.5	drop. 0.3	Gauss. 1.0	col. 0.5	bias 0.5	turb. mod.	turb. sev.
TECS autopilot	200	200	200	200	200	200	200
Plain MLP	200	200	200	200	200	200	200
Direct + MLP	177	197	–	–	–	–	–
Direct + physics (end to end)	200	200	200	200	200	200	200
JEPA + physics	200	200	200	200	200	200	200
JEPA + physics, context c	–	–	200	200	199	200	200
Direct frozen + physics	200	200	200	200	200	200	200

Survivors out of 200; – = not run. Gauss. = Gaussian, col. = colored, drop. = dropout, turb. = turbulence.

- **TECS**, the **Plain MLP** and every **physics-head** model keep **199–200 of 200** aircraft in every condition they flew (7 conditions; the context variants 5).
- **Direct + MLP** (flown in 2 conditions) loses aircraft: **177**/**200** under Gaussian 0.5 (McNemar $p < 0.001$), **197**/**200** under dropout. **JEPA + MLP** was not flown here; it crashed every aircraft in an earlier 40-aircraft test.
- With the best model (**JEPA + physics**), MPPI tracks airspeed **32 %** better than **TECS** under Gaussian 0.5 (**1.19** vs **1.75 m/s**); **TECS** has no sensor filter and tracks roll better.

Survival does not separate the **Plain MLP** and the physics-head models; tracking does.

8 Conclusions

1. SkyJEPA's ranking reproduces on training-range aircraft, but the **physics head** trained end to end (**5.8 m**) and a **Plain MLP** (**6.0 m**) beat **JEPA + physics** (**8.1 m**), and **JEPA + physics** is the worst on extreme aircraft.
2. Here the commands are not forces, so the model must identify the airframe; the **JEPA** latent keeps little motor and CG information (R^2 **0.09** / **0.30** vs **0.49** / **0.75**), even with a free context latent.
3. The **JEPA** trunk is less sensitive to sensor noise only when no model is trained with noise (**3–9 %**). With the same noise augmentation, **Plain MLP + noise** and **Direct + physics + noise** track **12–27 %** better than **JEPA + physics + noise** under every noise kind, including kinds never seen in training.
4. With 9 states, a **Plain MLP** of 10.6k parameters on the raw window predicts as well as the best physics model and better than the JEPA models.

Where JEPA could still help: rich observations with distractor channels or images (next step).

Limits: simulation only; one airframe family; one training seed per model in closed loop; physics heads receive the true mass and inertia, the other models do not; noise-trained models were selected on clean validation flights; the link to "commands are not forces" is not tested by varying the command type.

References

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Code and results: `https://github.com/clementdumeril/WingJEPA`